Week 4: Lecture and In-Class Problems

Tashkent University of Technology | Linear Algebra I | Fall 2026

Matrix rank and matrix inverse

80-minute lecture followed by 90 minutes of problems. The lecture includes a one-page Week 3 recap, rank, inverse matrices, and a guided introduction to Gauss-Jordan elimination.

Lecture timing: recap 5 min; rank and geometry 15; row operations and rank computation 15; inverse definition and formula 15; Gauss-Jordan examples 20; inverse rules and questions 10.

Practice timing: problems 1-5, 20 min; 6-9, 25 min; 10-12, 25 min; selected challenges 13-15, 20 min. Unfinished challenges can be continued after class.

All vectors use column notation. Every problem starts on a new PDF page. Statements and solutions are separated by generous blank space. The lecture also has a separate Beamer presentation with alternating problem and solution slides.

Week 3 recap: one page

A matrix product is defined when the inner dimensions match: $(m \times n)(n \times p)$ gives an $m \times p$ matrix. The columns of AB are A applied to the columns of B. In $AB\mathbf{x}$, B acts first.

$$(AB)^T = B^T A^T, \qquad AB \neq BA \text{ in general.}$$

For a square matrix, the determinant is a scalar. In two dimensions,

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc.$$

The absolute determinant measures area or volume scaling. A zero determinant means that full-dimensional area or volume has collapsed. We will now connect this loss of information to rank and invertibility.

$$\det(AB) = \det A \det B, \qquad \det(A^T) = \det A.$$

For three columns $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^3$,

$$\det[\mathbf{u}\ \mathbf{v}\ \mathbf{w}] = (\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}.$$

Its absolute value is parallelepiped volume. Keep every vector in column form, and distinguish a matrix from its scalar determinant.

1. Rank counts independent directions

The **rank** of an $m \times n$ matrix is the largest number of linearly independent columns. Equivalently, it is the dimension of the space spanned by the columns. The number of independent rows is the same.

$$0 \leq \operatorname{rank} A \leq \min(m, n).$$

Since $A\mathbf{x}$ is a combination of columns, rank counts the independent directions available in the output. Rank zero means that every entry is zero. A nonzero matrix can have rank one even if every entry is nonzero.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}, \qquad \operatorname{rank} A = 1.$$

Every column is a multiple of $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$, and the first column is nonzero. Counting nonzero columns would give the wrong answer.

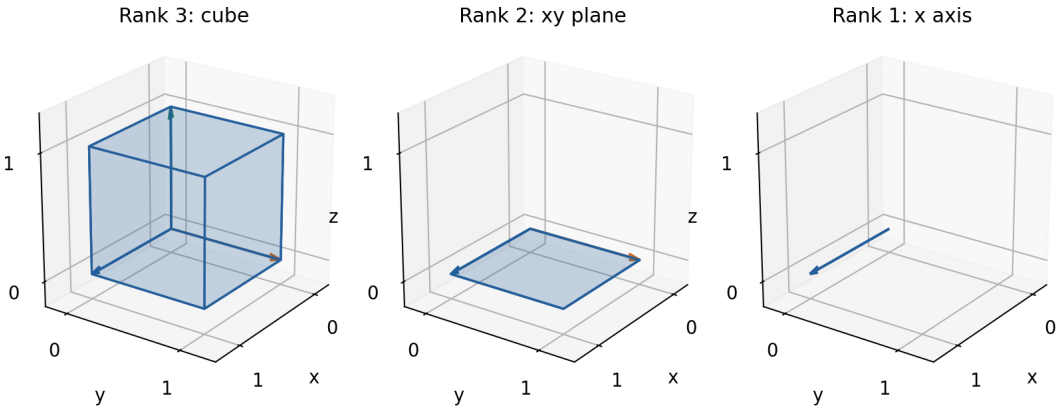
2. Rank in three-dimensional Euclidean space

Apply each of the following matrices to columns in $\mathbb{R}^3$:

$$I_3, \qquad P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

I_3 preserves all three coordinates and has rank 3. P discards z and has rank 2; its outputs lie in the xy plane. Q retains only x and has rank 1; its outputs lie on the x axis.

The figure shows images of the unit cube. The complete output spaces are the whole three-dimensional space, the whole plane, and the whole line, respectively. A bounded cube is used only to make the action visible.



3. A rank-two matrix can have three nonzero columns

Consider

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix} = [\mathbf{a}_1 \ \mathbf{a}_2 \ \mathbf{a}_3].$$

The third column satisfies $\mathbf{a}_3 = \mathbf{a}_1 + \mathbf{a}_2$. The first two are independent: if $c_1\mathbf{a}_1 + c_2\mathbf{a}_2 = \mathbf{0}$, the first two coordinates give $c_1 = c_2 = 0$. Thus the rank is exactly 2.

Every output satisfies $z = x + y$, so the image is a plane through the origin. The determinant is zero, but the rank is not zero. A zero determinant alone does not distinguish ranks 0, 1, and 2 for a 3×3 matrix.

4. Row operations preserve rank

Three reversible operations are allowed:

1. Exchange two rows: $R_i \leftrightarrow R_j$.
2. Multiply a row by a nonzero scalar: $R_i \leftarrow cR_i$.
3. Add a multiple of another row: $R_i \leftarrow R_i + cR_j$.

These operations preserve rank because each can be undone. They preserve linear relations among columns, although they generally change the actual column vectors and the space they occupy.

Multiplying a row by zero is not allowed: it can destroy information. When working with an augmented matrix, operate on the **entire row**, including the entries to the right of the dividing line. We introduce these operations here to compute rank and inverses; the systematic solution of general linear systems follows in Week 5.

5. Echelon form and pivots

In row echelon form, all zero rows are at the bottom and the first nonzero entry of each nonzero row lies to the right of the leading entry in the row above. These leading entries are the **pivots**; entries beneath a pivot are zero.

$$\begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 5 & 1 & 3 \\ 3 & 7 & 1 & 4 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 - 3R_1$, and then $R_3 \leftarrow R_3 - R_2$. There are two pivots, so the rank is 2.

The first and second columns are pivot columns. To select independent columns of the original matrix, take columns 1 and 2 of the **original** matrix. Counting nonzero rows gives the rank only after reaching echelon form.

6. Rank from determinants of submatrices

A $k \times k$ minor is the determinant of a square submatrix obtained by choosing k rows and k columns. The rank is the largest size of a nonzero minor.

For a 3×3 matrix, a nonzero determinant gives rank 3. If the determinant is zero but some 2×2 minor is nonzero, the rank is 2. If all 2×2 minors vanish but some entry is nonzero, the rank is 1.

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}, \qquad \det A = 0, \qquad \det \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 1.$$

Therefore $\operatorname{rank} A = 2$. The nonzero minor gives a lower bound; the vanishing full determinant gives the upper bound. For larger matrices, row reduction is usually more efficient than checking many minors.

7. The inverse undoes a matrix

For a square matrix A, an inverse is a matrix A^{-1} of the same size satisfying

$$A^{-1}A = I = AA^{-1}.$$

If $\mathbf{y} = A\mathbf{x}$, multiplication by A^{-1} recovers the input: $\mathbf{x} = A^{-1}\mathbf{y}$. An inverse is not obtained by taking the reciprocal of each entry.

The inverse, if it exists, is unique. If B and C are both inverses, then $B = B(AC) = (BA)C = C$. A matrix with an inverse is **invertible** or **nonsingular**; a square matrix without one is **singular**.

An ordinary two-sided inverse is defined here for square matrices. Rectangular matrices can have one-sided inverses, which we explore only as a challenge.

8. When does the inverse exist?

For an $n \times n$ matrix A, the following conditions are equivalent:

$$A^{-1} \text{ exists} \quad \Longleftrightarrow \quad \operatorname{rank} A = n \quad \Longleftrightarrow \quad \det A \neq 0.$$

They also mean that row reduction produces n pivots and that $A\mathbf{x} = \mathbf{0}$ has only the zero solution.

If A sends some nonzero $\mathbf{z}$ to zero, no inverse can recover that input: $\mathbf{z} = A^{-1}A\mathbf{z} = A^{-1}\mathbf{0} = \mathbf{0}$ would be a contradiction. Thus invertibility requires that no direction be lost.

For rectangular matrices, full rank means rank $\min(m, n)$. Full rank alone does not create an ordinary two-sided inverse when $m \neq n$.

9. The 2 by 2 inverse formula

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $\Delta = ad - bc$. If $\Delta \neq 0$,

$$A^{-1} = \frac{1}{\Delta} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Exchange the diagonal entries, change the signs of the off-diagonal entries, and divide the entire matrix by the determinant. The formula follows because multiplying A by the numerator matrix gives ΔI.

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \quad \Delta = 1, \quad A^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}.$$

For example, the top-left entry of AA^{-1} is $2 - 1 = 1$, and its top-right entry is $-2 + 2 = 0$. The other row gives 0 and 1. If $\Delta = 0$, division is impossible and the matrix is singular.

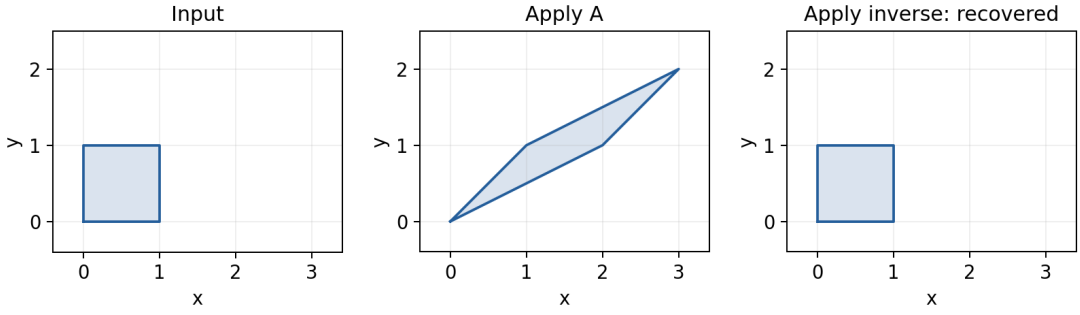
10. Geometry: transformation and recovery

For the explicit pair

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \qquad B = A^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix},$$

the unit square is mapped to the parallelogram with edge columns $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Applying B to every point of this image recovers the original square.

The area is unchanged because $\det A = 1$, but the shape changes. Area preservation does not mean that A is the identity or that it preserves angles. The coordinates in all three panels use the same scale.



11. Gauss-Jordan: augment with the identity

To compute the inverse, apply the same row operations to A and I:

$$[A \mid I] \longrightarrow [I \mid A^{-1}].$$

Why does this work? Each row operation is left multiplication by a reversible matrix. If all the operations together are represented by E, then $EA = I$. Applying the same operations to the right block produces $EI = E = A^{-1}$.

Choose a nonzero pivot, swapping rows when necessary. Scale the pivot to 1, then eliminate the other entries in its column. Repeat until the left block is the identity. If the left block cannot acquire a pivot in every column, the inverse does not exist.

The dividing line in the augmented matrix is only a visual separator. Operations must act on both sides.

12. A complete 2 by 2 Gauss-Jordan example

Start with $A = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$ and append the identity:

$$\left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 2 & 3 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{array}\right].$$

Now clear the entry above the second pivot:

$$\xrightarrow{R_1 \leftarrow R_1 - R_2} \left[\begin{array}{cc|cc} 1 & 0 & 3 & -1 \\ 0 & 1 & -2 & 1 \end{array}\right].$$

Hence $A^{-1} = \begin{bmatrix} 3 & -1 \\ -2 & 1 \end{bmatrix}$. Both row operations were applied to all four entries of the affected row. Check by multiplication before accepting an inverse.

13. A 3 by 3 inverse and the importance of order

Consider an upper triangular matrix with unit diagonal:

$$U = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}.$$

Start from $[U \mid I_3]$. First use $R_2 \leftarrow R_2 - 3R_3$. Then use $R_1 \leftarrow R_1 - 2R_2$, with the **updated** row 2. This gives

$$[I_3 \mid U^{-1}], \qquad U^{-1} = \begin{bmatrix} 1 & -2 & 6 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix}.$$

The 6 in the upper-right corner is essential: it cancels the two-step coupling through the second coordinate. Negating the two off-diagonal entries alone would not give an inverse. This example needs only two row operations because the matrix already has a simple structure.

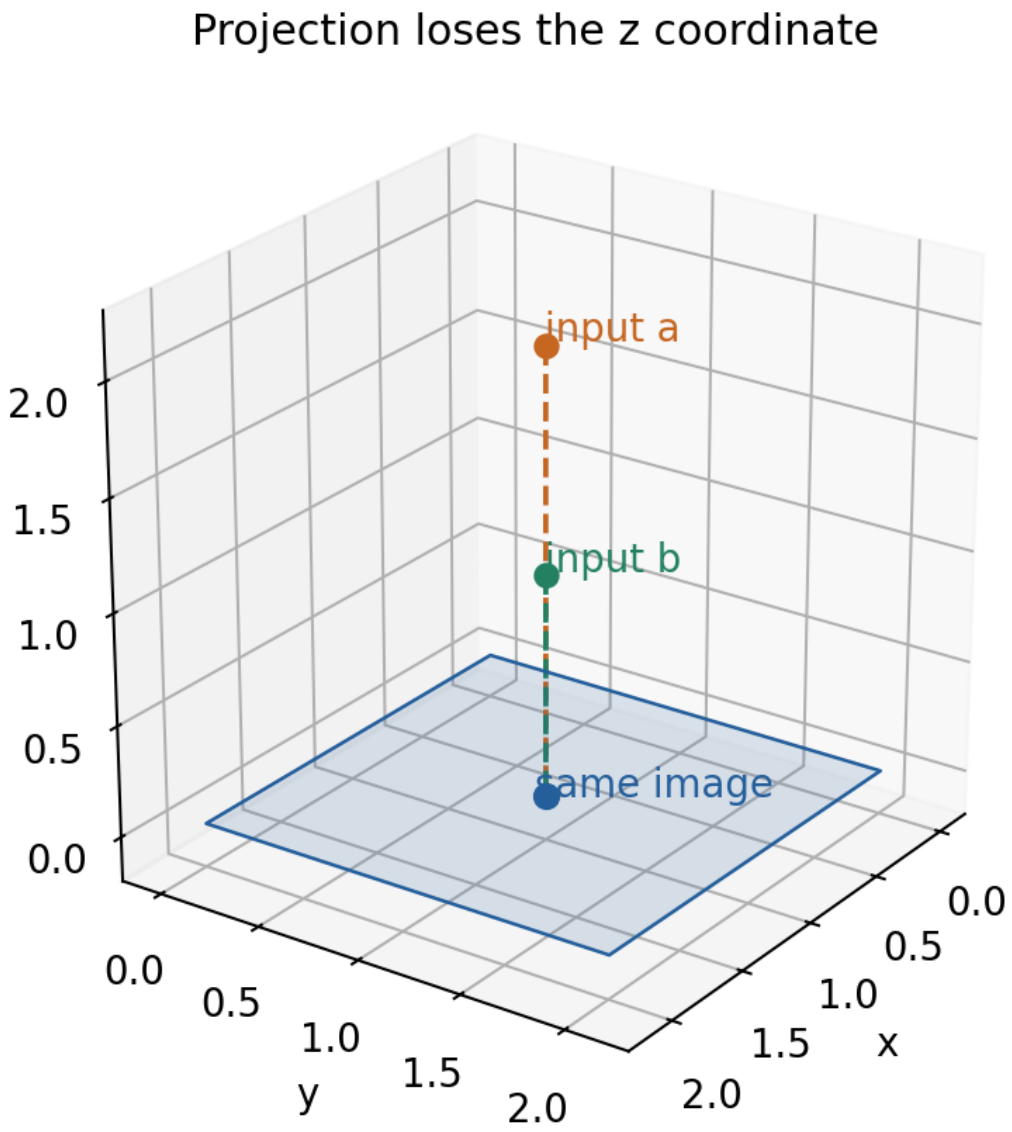
14. Why a singular map cannot be reversed

For the projection

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad P \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = P \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

Two distinct input columns have the same output. An inverse would have to send that output back to both inputs, which a function cannot do. The rank is 2 and the determinant is zero.

More generally, if $A\mathbf{z} = \mathbf{0}$ for a nonzero $\mathbf{z}$, then $A(\mathbf{x} + \mathbf{z}) = A\mathbf{x}$. Adding a lost direction leaves the output unchanged.



15. Inverse rules: reverse the order

For invertible square matrices of the same size and a nonzero scalar c,

$$(AB)^{-1} = B^{-1}A^{-1}, \quad (A^T)^{-1} = (A^{-1})^T,$$

$$(cA)^{-1} = c^{-1}A^{-1}, \quad (A^{-1})^{-1} = A.$$

To undo AB, undo A first and then B. Algebraically, $(AB)(B^{-1}A^{-1}) = AIA^{-1} = I$. The reverse multiplication also gives I.

There is no general rule $(A+B)^{-1} = A^{-1} + B^{-1}$. Taking $A = B = I$ already disproves it: the left side is $\frac{1}{2}I$, while the right side is $2I$. Check that the sum is invertible before attempting its inverse.

16. Solving a small system using the inverse

If A is invertible, the unique solution of $A\mathbf{x} = \mathbf{b}$ is $\mathbf{x} = A^{-1}\mathbf{b}$. For example,

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Substitute into the original equation: $2(2) + 1 = 5$ and $2 + 1 = 3$. This check can catch arithmetic or order errors.

For large numerical systems, solving directly by elimination is usually preferable to explicitly forming an inverse. Here, computing inverses helps us understand rank, reversibility, and uniqueness. General system-solving methods come next week.

Reading and transition to practice

Reading: Gilbert Strang, *Introduction to Linear Algebra*, supplied 4th edition, §2.5 (inverse matrices and Gauss-Jordan) and the introductory rank-and-pivot material in §3.3. The rank-by-minors test connects with the determinant material in Chapter 5. Week 4 of the updated syllabus specifies matrix rank and inverse; later weeks develop elimination, null spaces, and fundamental subspaces in more depth.

Before practice, explain the difference between matrix size and rank; why rank is unchanged by reversible row operations; why a square matrix needs full rank to be invertible; and why an augmented row operation acts on both blocks.

Keep an exact fraction until the final answer. Check every proposed inverse by multiplication, and explain a failure to invert using rank or a nonzero vector mapped to zero.

Part II: In-class problems

The theory section ends here. Take a short pause before starting the problem session.

Begin with rank and simple inverses, then move to Gauss-Jordan calculations and applications. The final three problems are challenges. Try each problem before reading its solution.

Problem 1: Ranks by inspection - basic

Find the rank of each matrix and give a reason:

$$Z = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}, \qquad D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}.$$

Solution 1

Z has no independent nonzero column, so its rank is 0. In B, column 2 is 2 times column 1, and column 1 is nonzero; its rank is 1. The two columns of D point along different coordinate axes, so its rank is 2.

Problem 2: Independent and dependent columns - basic

Let $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 1 & 1 & 3 \end{bmatrix}$. Express column 3 as a combination of columns 1 and 2. Determine the rank and describe the plane containing every output $A\mathbf{x}$.

Solution 2

Column 3 equals 2 times column 1 plus column 2. The first two columns are independent because their first two coordinates are $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Therefore the rank is 2. Each column satisfies $z = x + y$, so every output lies in that plane.

Problem 3: Rank by row reduction - basic

Reduce $C = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 5 & 1 & 3 \\ 3 & 7 & 1 & 4 \end{bmatrix}$ to echelon form. State its rank and select independent columns from the original matrix.

Solution 3

Use $R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 - 3R_1$, then $R_3 \leftarrow R_3 - R_2$:

$$C \longrightarrow \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There are two pivots, in columns 1 and 2. The rank is 2, and columns 1 and 2 of the original C are an independent pair.

Problem 4: Confirm rank with a minor - basic

For $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 1 & 1 & 3 \end{bmatrix}$, use its determinant and a 2×2 minor to establish the exact rank. Why is the full determinant alone insufficient?

Solution 4

The third column is a combination of the first two, so the full determinant is zero and rank is at most 2. The upper-left 2×2 submatrix is I_2, whose determinant is 1. Thus rank is at least 2, and therefore exactly 2. A zero 3×3 determinant permits ranks 0, 1, or 2.

Problem 5: Diagonal inverse - basic

Find the inverse of $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$. State what happens to its rank and inverse if its last diagonal entry is replaced by zero.

Solution 5

Undo each coordinate scaling:

$$D^{-1} = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & -\frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{3} \end{bmatrix}.$$

The diagonal products are all 1, so $DD^{-1} = I_3$. If the final entry is zero, the rank drops to 2 and the third coordinate is lost; no inverse exists.

Problem 6: A 2 by 2 inverse - intermediate

Find A^{-1} for $A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ using the determinant formula. Verify your answer by computing AA^{-1}.

Solution 6

The determinant is $2(1) - (1)(1) = 1$. Therefore

$$A^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}, \qquad \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

The nonzero determinant also confirms that the rank is 2.

Problem 7: Recover an input column - intermediate

Solve $A\mathbf{x} = \mathbf{b}$ using the inverse, where

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

Check your solution in the original equation.

Solution 7

Use the inverse from the 2×2 formula:

$$\mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

Substitution gives $A \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$, as required. Because A is invertible, this solution is unique.

Problem 8: Gauss-Jordan with a row swap - intermediate

Use Gauss-Jordan elimination to invert $H = \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix}$. Show the augmented matrix after the row swap and at the end.

Solution 8

Swap rows 1 and 2, then replace R_1 by $R_1 - 2R_2$:

$$\left[\begin{array}{cc|cc} 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{array}\right] \longrightarrow \left[\begin{array}{cc|cc} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{array}\right].$$

$$\longrightarrow \left[\begin{array}{cc|cc} 1 & 0 & -2 & 1 \\ 0 & 1 & 1 & 0 \end{array}\right].$$

Thus $H^{-1} = \begin{bmatrix} -2 & 1 \\ 1 & 0 \end{bmatrix}$. A zero leading entry required a swap, not a declaration that the matrix was singular.

Problem 9: Gauss-Jordan for a 3 by 3 matrix - intermediate

Find the inverse of $U = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$ by applying row operations to $[U \mid I_3]$. Verify the upper-right entry of UU^{-1}.

Solution 9

Use $R_2 \leftarrow R_2 - 2R_3$, then $R_1 \leftarrow R_1 - 2R_2$ with the updated row 2. The right block changes as follows:

$$I_3 \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \longrightarrow U^{-1} = \begin{bmatrix} 1 & -2 & 4 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}.$$

The left block becomes I_3. The upper-right entry of UU^{-1} is $4 + 2(-2) = 0$, as required.

Problem 10: Inverse of a product - intermediate

Let $S = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and $D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$. Find $(SD)^{-1}$ without applying the 2×2 formula directly to SD. Check the order of the factors.

Solution 10

Reverse the factor order:

$$(SD)^{-1} = D^{-1}S^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -1 \\ 0 & \frac{1}{3} \end{bmatrix}.$$

Indeed, $(SD)(D^{-1}S^{-1}) = SIS^{-1} = I$. The other order gives $S^{-1}D^{-1} = \begin{bmatrix} \frac{1}{2} & -\frac{2}{3} \\ 0 & \frac{1}{3} \end{bmatrix}$, a different matrix.

Problem 11: Prove that an inverse does not exist - intermediate

Let $B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$. Find a nonzero column $\mathbf{z}$ with $B\mathbf{z} = \mathbf{0}$. Use it to explain why B has no inverse. State its rank.

Solution 11

Take $\mathbf{z} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$. Then

$$B\mathbf{z} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

If an inverse existed, multiplication by it would force $\mathbf{z} = B^{-1}\mathbf{0} = \mathbf{0}$, a contradiction. Since column 2 is 2 times the nonzero first column, the rank is 1.

Problem 12: Rank and information loss in xyz space - intermediate

Let $P = \text{diag}(1, 1, 0)$ and $L = \text{diag}(1, 2, 2)$. Find their ranks. Compute the images of $\mathbf{a} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ under P. Which transformation can be reversed?

Solution 12

P has rank 2, and

$$P\mathbf{a} = P\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}.$$

It loses the third coordinate, so it cannot be reversed. L has rank 3 and inverse $\text{diag}(1, \frac{1}{2}, \frac{1}{2})$. Its three coordinate scale factors are all nonzero.

Problem 13: Rank changes with a parameter - challenge

For

$$A(k) = \begin{bmatrix} 1 & 1 & 0 \\ 0 & k-2 & 0 \\ 0 & 0 & k+1 \end{bmatrix},$$

find the rank for every real k. For which k does the inverse exist? Find that inverse when it exists.

Solution 13

The determinant is $(k-2)(k+1)$. The rank is 3 except at $k = 2$ or $k = -1$; at each of those values exactly two independent columns remain, so the rank is 2. For $k \neq 2, -1$,

$$A(k)^{-1} = \begin{bmatrix} 1 & -\frac{1}{k-2} & 0 \\ 0 & \frac{1}{k-2} & 0 \\ 0 & 0 & \frac{1}{k+1} \end{bmatrix}.$$

Multiplication cancels the upper off-diagonal entry and gives I_3.

Problem 14: A one-sided inverse - challenge

Let $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$. Compute AB and BA. Does B qualify as an ordinary two-sided inverse of A? State the ranks of A and B.

Solution 14

The products are

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad BA = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

$AB = I_2$, but $BA \neq I_3$. Thus B is a right inverse of A, not a two-sided inverse. Both matrices have rank 2; full rank for rectangular matrices does not imply a two-sided inverse.

Problem 15: Solve a matrix equation - challenge

Find the matrix X satisfying $AXB = C$, where

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & 9 \\ 1 & 5 \end{bmatrix}.$$

Show which side you multiply on at each step.

Solution 15

Multiply on the left by A^{-1}, then on the right by B^{-1}:

$$X = A^{-1}CB^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 9 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}.$$

The order matters because matrix multiplication is not generally commutative. Substitution gives $A \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} B = C$.